Supplemental Materials

Content

Supplemental Material I **3**

Table S1. PRISMA Checklist 3

Supplementary Material II – Search strategies **6**

PubMed 6

Embase 6

Cochrane Central Register of Controlled Trials (CENTRAL) 6

Clinical Trials.gov 7

Supplemental Material III **8**

Table S2. Quality of cohort studies included in the systematic review and meta-analysis 8

Supplemental Material IV **10**

Table S3. Pooled results of the association between *RFC1 (rs1051266)* and HDMTX related outcomes 10

Table S4. Pooled results of the association between *SLCO1B1 (rs4149056)* and HDMTX related outcomes 11

Table S5. Pooled results of the association between *ABCB1 (rs1045642)* and HDMTX related outcomes 12

Table S6. Pooled results of the association between *FPGS (rs10106)* and HDMTX related outcomes 13

Table S7. Meta-analysis results of the association between *FPGS (rs1544105)* and HDMTX related outcomes 13

Table S8. Pooled results of the association between *MTHFR (rs1801133)* and HDMTX related outcomes 14

Table S9. Pooled results of the association between *MTHFR (rs1801131)* and HDMTX related outcomes 16

Table S10. Pooled results of the association between *TYMS (rs34743033)* and HDMTX related outcomes 18

Table S11. Pooled results of the association between *ATIC (rs2372536)* and HDMTX related outcomes 18

Supplemental Material V 19

Figure S1. Forest plot for meta-analysis of the association between *RFC1 (rs1051266)* and hepatotoxicity under recessive model 19

Figure S2. Forest plot for meta-analysis of the association between ABCB1 (rs1045642) and hepatotoxicity under dominant model 20

Figure S3. Forest plot for meta-analysis of the association between *MTHFR (rs1801133)* and hepatotoxicity under dominant model 21

Figure S4. Forest plot for meta-analysis of the association between *TYMS (rs34743033)* and mucositis under dominant model 22

Supplemental Material VI 23

Figure S5. Funnel plot for publication bias test of the association between *MTHFR (rs1801133)* and hepatotoxicity under dominant model 23

Supplemental Material I

Table S1. PRISMA Checklist

Section/Topic	Item	Checklist item	Page
Title			
Title	1	Identify the report as a systematic review, meta-analysis, or both	P1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable, background, objectives, data sources, study eligibility criteria, participants, interventions, study appraisal and synthesis methods, results, limitations, conclusions and implications of key findings, systematic review registration number	P2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known	P3-4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS)	P4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (such as web address), and, if available, provide registration information including registration number	P4
Eligibility criteria	6	Specify study characteristics (such as PICOS, length of follow-up) and report characteristics (such as years considered, language, publication status) used as criteria for eligibility, giving rationale	P4-5
Information sources	7	Describe all information sources (such as databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched	P5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated	Supple II
Study selection	9	State the process for selecting studies (that is, screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis)	P5
Data collection process	10	Describe method of data extraction from reports (such as piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators	P6

Section/Topic	Item	Checklist item	Page
Data items	11	List and define all variables for which data were sought (such as PICOS, funding sources) and any assumptions and simplifications made	P6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis	P6
Summary measures	13	State the principal summary measures (such as risk ratio, difference in means)	P6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (such as I^2 statistic) for each meta-analysis	P7
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (such as publication bias, selective reporting within studies)	P7
Additional analyses	16	Describe methods of additional analyses (such as sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified	P7
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram	P7
Study characteristics	18	For each study, present characteristics for which data were extracted (such as study size, PICOS, follow-up period) and provide the citations	P8
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome-level assessment (see item 12).	P8
Results of individual studies	20	For all outcomes considered (benefits or harms), present for each study (a) simple summary data for each intervention group and (b) effect estimates and confidence intervals, ideally with a forest plot	P8-12
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency	P8-12
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see item 15)	P12
Additional analysis	23	Give results of additional analyses, if done (such as sensitivity or subgroup analyses, meta-regression) (see item 16)	P9-12

Section/Topic	Item	Checklist item	Page
DISCUSSION			
Summary of evidence	24	Summarise the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (such as health care providers, users, and policy makers)	P12-13
Limitations	25	Discuss limitations at study and outcome level (such as risk of bias), and at review level (such as incomplete retrieval of identified research, reporting bias)	P15-16
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research	P17-18
Funding			
Funding	27	Describe sources of funding for the systematic review and other support (such as supply of data) and role of funders for the systematic review	P18

Supplementary Material II – Search strategies

PubMed

#1 Methotrexate[MeSH] OR Methotrexate[Title/Abstract] OR MTX[Title/Abstract] OR Mexate[Title/Abstract] OR Amethopterin[Title/Abstract]
#2 "Polymorphism, single nucleotide"[MeSH Terms] OR "Pharmacogenetics"[MeSH Terms] OR "Single nucleotide polymorphism"[Title/Abstract] OR "SNP"[Title/Abstract] OR "Polymorphisms"[Title/Abstract] OR "Pharmacogenetic"[Title/Abstract] OR "Pharmacogenomics"[Title/Abstract] OR "Genetic"[Title/Abstract] OR "Genomic"[Title/Abstract] OR "Genotype"[Title/Abstract] OR "Gene"[Title/Abstract]
#3 "Hematologic neoplasms"[MeSH] OR "Leukemia"[MeSH] OR "Lymphoma"[MeSH] OR Hematologic neoplasm[Title/Abstract] OR Hematological neoplasm[Title/Abstract] OR Hematologic malignancy[Title/Abstract] OR Hematological malignancy[Title/Abstract] OR Leukemia[Title/Abstract] OR Lymphoma[Title/Abstract]
#4 Osteosarcoma[MeSH] OR Osteosarcoma[Title/Abstract] OR Sarcoma[Title/Abstract]

#1 AND #2 AND (#3 OR #4)
Filter: Human
Up to 2020.11.11

Embase

#1 'Methotrexate'/exp OR 'Methotrexate':ab,ti OR 'MTX':ab,ti OR 'Mexate':ab,ti OR 'Amethopterin':ab,ti
#2 'single nucleotide polymorphism'/exp OR 'pharmacogenetics'/exp OR ‘Single nucleotide polymorphism’:ab,ti OR ‘SNP’:ab,ti OR ‘Polymorphisms’:ab,ti OR ‘Pharmacogenetic’:ab,ti OR ‘Pharmacogenomics’:ab,ti OR ‘Genetic’:ab,ti OR ‘Genomic’:ab,ti OR ‘Genotype’:ab,ti OR ‘Gene’:ab,ti
#3 'hematologic malignancy'/exp OR 'leukemia'/exp OR 'lymphoma'/exp OR 'Hematologic neoplasm':ab,ti OR 'Hematological neoplasm':ab,ti OR 'Hematologic malignancy':ab,ti OR 'Hematological malignancy':ab,ti OR 'Leukemia':ab,ti OR 'Lymphoma':ab,ti
#4 'osteosarcoma'/exp OR 'Osteosarcoma':ab,ti OR 'Sarcoma':ab,ti

#1 AND #2 AND (#3 OR #4)
Filter: Human
Up to 2020.11

Cochrane Central Register of Controlled Trials (CENTRAL)

#1 Methotrexate[MeSH] OR Methotrexate:ti,ab,kw OR MTX:ti,ab,kw OR Mexate:ti,ab,kw OR Amethopterin:ti,ab,kw
#2 "Polymorphism, single nucleotide"[MeSH Terms] OR "Pharmacogenetics"[MeSH Terms] OR ("Single nucleotide polymorphism" OR "SNP" OR "Polymorphisms" OR "Pharmacogenetic" OR

"Pharmacogenomics" OR "Genetic" OR "Genomic" OR "Genotype" OR "Gene"):ti,ab,kw
#3 "Hematologic neoplasms"[MeSH] OR "Leukemia"[MeSH] OR "Lymphoma"[MeSH] OR (Hematologic neoplasm OR Hematological neoplasm OR Hematologic malignancy OR Hematological malignancy OR Leukemia OR Lymphoma):ti,ab,kw
#4 Osteosarcoma[MeSH] OR (Osteosarcoma OR Sarcoma):ti,ab,kw

#1 AND #2 AND (#3 OR #4)
Up to 2020.11.11

Clinical Trials.gov

Methotrexate OR MTX | Hematologic Malignancy OR osteosarcoma
Up to 2020.11.11

Supplemental Material III

Table S2. Quality of cohort studies included in the systematic review and meta-analysis

Study	Selection of cohorts				Comparability of cohorts	Assessment of outcome			Score
	Representativeness of exposed cohort	Selection of non-exposed cohort	Ascertainment of exposure	Outcome was not present at start of study	Comparability	Assessment of outcome	Long enough follow-up	Adequacy of follow-up	
Esmaili M A-2020	*	*	*		**	*		*	7
Kotur N-2020	*	*	*		**	*	*		7
Liu S G-2017	*	*	*		**	*	*		7
Den Hoed M-2014	*	*	*	*	**	*	*		8
Suthandiram S-2014	*	*	*		**	*	*	*	8
Yanagimachi M-2013	*	*	*		**		*		6
Chiusolo P-2012	*	*	*	*	**	*	*		8
Faganel K B-2011	*	*	*		**	*	*	*	8
Faganel K B-2010	*	*	*		**	*	*	*	8
Ashton L J-2009	*	*	*		**	*	*	*	8
Imanishi H-2007	*	*	*		**	*	*	*	8
Shimasaki N-2006	*	*	*		**	*	*	*	8
Kishi S-2003	*	*	*		*	*	*	*	7
Laverdiere C-2002	*	*	*		**		*	*	7
Yang L-2017	*	*	*		**	*	*	*	8
Avivi I-2014	*	*	*	*	**	*	*	*	9
Fukushima H-2013	*	*	*		**	*	*	*	8
Tsujimoto S-2016	*	*	*		**	*	*	*	8

Study	Selection of cohorts				Comparability of cohorts	Assessment of outcome			Score
	Representativeness of exposed cohort	Selection of non-exposed cohort	Ascertainment of exposure	Outcome was not present at start of study	Comparability	Assessment of outcome	Long enough follow-up	Adequacy of follow-up	
Ma C X-2015	*	*	*		**	*	*	*	8
Chae H-2020	*	*	*		**	*	*	*	8
Chang X-2021	*	*	*		**	*	*	*	8
Giletti-2017	*	*	*		**	*	*	*	7
Yaz c o-2017	*	*	*		**	*	*		7
Choi Y J-2016	*	*	*		**	*	*	*	8
Erculj N-2014	*	*	*		**	*	*	*	8
Erculj, N-2012	*	*	*		**	*	*	*	8
Haase R-2012	*	*	*		**	*	*		7
D'Angelo V-2011	*	*	*		**	*	*		7
Liu S G-2011	*	*	*		**	*	*	*	8
Ruiz-Arguelles G J-2007	*	*	*		*		*	*	6
Seidemann-2006	*	*	*		**	*	*		7
Huang Z-2016	*	*	*	*	**	*	*	*	9
Oosterom-2018	*	*	*		**	*	*		7
Radtke-2013	*	*	*		**	*	*	*	8

Supplemental Material IV

Table S3. Pooled results of the association between *RFC1 (rs1051266)* and HDMTX related outcomes

RFC1 A80G (rs1051266)	MM/Mm vs mm （dominant model)					MM vs Mm/mm (recessive model)					M vs m (allelic model)				
	Studies	OR or HR (95%CI)	*Phet*	I^2	P-sub	Studies	OR or HR (95%CI)	*Phet*	I^2	P-sub	Studies	OR or HR (95%CI)	*Phet*	I^2	P-sub
Hepatotoxicity	5	0.62 [0.33, 1.16]	0.02	65%	—	**3**	**0.35 [0.16, 0.76]**	**0.91**	**0%**	—	4	0.77 [0.48, 1.21]	0.68	0%	—
Pediatric	3	0.82 [0.32, 2.07]	0.01	78%	NS [a]	1	0.28 [0.06, 1.38]	NA [b]	NA	NS	2	0.97 [0.51, 1.84]	0.68	0%	NS
Adult	2	0.49 [0.21, 1.15]	0.22	34%		**2**	**0.39 [0.16, 0.93]**	**0.79**	**0%**		2	0.60 [0.31, 1.16]	0.57	0%	
G3-4 hepatotoxicity *(Pediatric)*	1	0.38 [0.04, 3.29]	NA	NA	—	—	—	—	—	—	—	—	—	—	—
Renal toxicity	2	0.93 [0.41, 2.10]	0.88	0%	—	2	0.73 [0.21, 2.56]	0.38	0%	—	2	0.91 [0.49, 1.70]	0.71	0%	—
Pediatric	1	0.96 [0.37, 2.50]	NA	NA	NS	1	1.08 [0.23, 5.00]	NA	NA	NS	1	0.98 [0.48, 2.00]	NA	NA	NS
Adult	1	0.83 [0.17, 4.04]	NA	NA		1	0.32 [0.04, 2.91]	NA	NA		1	0.74 [0.21, 2.64]	NA	NA	
Mucositis	4	0.91 [0.54, 1.52]	1	0%	—	3	0.99 [0.60, 1.61]	0.94	0%	—	4	0.90 [0.61, 1.32]	0.61	0%	—
Pediatric	3	0.95 [0.51, 1.74]	1	0%	NS	2	1.03 [0.60, 1.78]	0.99	0%	NS	3	0.93 [0.60, 1.44]	0.41	0%	NS
Adult	1	0.83 [0.32, 2.15]	NA	NA		1	0.81 [0.26, 2.55]	NA	NA		1	0.81 [0.37, 1.79]	NA	NA	
G3-4 Mucositis *(Pediatric)*	2	1.17 [0.54, 2.52]	0.23	30%	—	3	0.95 [0.53, 1.70]	0.59	0%	—	2	0.97 [0.58, 1.62]	0.49	0%	—
Neurotoxicity *(Pediatric)*	2	1.22 [0.46, 3.28]	0.73	0%	—	2	1.08 [0.51, 2.27]	0.71	0%	—	2	1.07 [0.58, 2.00]	0.91	0%	—
G3-4 Neurotoxicity *(Pediatric)*	2	0.52 [0.16, 1.66]	0.8	0%	—	2	0.74 [0.22, 2.50]	0.38	0%	—	2	0.77 [0.31, 1.89]	0.69	0%	—
GI toxicity *(Pediatric)*	1	0.73 [0.29, 1.79]	NA	NA	—	—	—	—	—	—	1	3.14 [1.08, 9.09]	NA	NA	—
Overall toxicity *(Pediatric)*	1	0.65 [0.29, 1.49]	NA	NA	—	**1**	**0.34 [0.12, 0.91]**	**NA**	**NA**	—	1	0.70 [0.39, 1.26]	NA	NA	—
Therapeutic interference *(Pediatric)*	1	0.94 [0.32, 2.79]	NA	NA	—	1	2.15 [0.63, 7.35]	NA	NA	—	1	1.13 [0.55, 2.34]	NA	NA	—
Relapse *(Pediatric)*	1	3.52 [0.20, 62.83]	NA	NA	—	1	2.55 [0.56, 11.67]	NA	NA	—	1	1.89 [0.48, 7.43]	NA	NA	—
2y-PFS *(Adult)*	—	—	—	—	—	**1**	**3.30 [1.21, 9.03]**	**NA**	**NA**	—	—	—	—	—	—
2y-OS *(Adult)*	—	—	—	—	—	**1**	**2.87 [1.08, 7.65]**	**NA**	**NA**	—	—	—	—	—	—

RFC1 A80G (rs1051266)	MM/Mm vs mm （dominant model)					MM vs Mm/mm (recessive model)					M vs m (allelic model)				
	Studies	OR or HR (95%CI)	*Phet*	I^2	P-sub	Studies	OR or HR (95%CI)	*Phet*	I^2	P-sub	Studies	OR or HR (95%CI)	*Phet*	I^2	P-sub
5y-EFS *(Pediatric)*	—	—	—	—	—	2	1.43 [0.89, 2.30]	0.003	88%		—	—	—	—	—

Note: —: No available data

Abbreviation: a. NS: No statistical difference; b. NA: Not applicable

Table S4. Pooled results of the association between *SLCO1B1 (rs4149056)* and HDMTX related outcomes

SLCO1B1 T521C (rs4149056)	MM/Mm vs mm （dominant model)					MM vs Mm/mm (recessive model)					M vs m (allelic model)				
	Studies	OR or HR (95%CI)	*Phet*	I^2	P-sub	Studies	OR or HR (95%CI)	*Phet*	I^2	P-sub	Studies	OR or HR (95%CI)	*Phet*	I^2	P-sub
Hepatotoxicity	2	0.80 [0.44, 1.46]	0.0008	91%	—	—	—	—	—	—	—	—	—	—	—
Pediatric	1	0.31 [0.13, 0.76]	NA	NA	0.0009*	—	—	—	—	—	—	—	—	—	—
Adult	1	3.05 [1.12, 8.32]	NA	NA		—	—	—	—		—	—	—	—	
Renal toxicity *(Pediatric)*	1	1.02 [0.36, 2.91]	NA [a]	NA	—	—	—	—	—	—	—	—	—	—	—
G3-4 Mucositis *(Pediatric)*	2	0.48 [0.22, 1.03]	0.93	0%	—	1	0.56 [0.03, 10.15]	NA	NA	—	1	0.53 [0.20, 1.39]	NA	NA	—
GI toxicity *(Pediatric)*	—	—	—	—	—	1	1.65 [0.27, 10.02]	NA	NA	—	—	—	—	—	—
Overall toxicity *(Adult)*	1	0.41 [0.15, 1.15]	NA	NA	—	1	0.11 [0.01, 2.26]	NA	NA	—	1	0.50 [0.19, 1.33]	NA	NA	—
Relapse *(Pediatric)*	1	1.69 [0.50, 5.68]	NA	NA	—	—	—	—	—	—	—	—	—	—	—
5y-EFS *(Pediatric)*	—	—	—	—	—	1	12.24 [3.94, 38.07]	NA	NA	—	—	—	—	—	—

Note: —: No available data

Abbreviation: a. NA: Not applicable

Table S5. Pooled results of the association between *ABCB1 (rs1045642)* and HDMTX related outcomes

ABCB1 C3435T (rs1045642)	MM/Mm vs mm （dominant model)					MM vs Mm/mm (recessive model)					M vs m (allelic model)				
	Studies	OR or HR (95%CI)	*Phet*	I^2	P-sub	Studies	OR or HR (95%CI)	*Phet*	I^2	P-sub	Studies	OR or HR (95%CI)	*Phet*	I^2	P-sub
Hepatotoxicity	**3**	**3.80 [1.68, 8.61]**	**0.29**	**18%**	—	2	1.91 [0.89, 4.09]	0.17	48%	—	2	1.61 [0.89, 2.90]	0.26	21%	—
Pediatric	1	1.91 [0.60, 6.03]	NA [a]	NA	NS [b]	1	1.11 [0.37, 3.27]	NA	NA	NS	1	1.21 [0.55, 2.63]	NA	NA	NS
Adult	**2**	**7.46 [2.19, 25.39]**	**0.69**	**0%**		**1**	**3.38 [1.07, 10.68]**	**NA**	**NA**		1	2.40 [0.97, 5.92]	NA	NA	
Renal toxicity *(Adult)*	—	—	—	—	—	1	0.71 [0.19, 2.62]	NA	NA	—	—	—	—	—	—
Mucositis	3	0.82 [0.48, 1.41]	0.19	39%	—	2	0.79 [0.41, 1.54]	0.24	27%	—	2	1.01 [0.63, 1.62]	0.74	0%	—
Pediatric	1	1.56 [0.57, 4.27]	NA	NA	NS	1	0.53 [0.19, 1.44]	NA	NA	NS	1	0.95 [0.53, 1.71]	NA	NA	NS
Adult	2	0.60 [0.31, 1.15]	0.38	0%		1	1.19 [0.47, 3.04]	NA	NA		1	1.13 [0.50, 2.54]	NA	NA	
G3-4 Mucositis *(Pediatric)*	1	1.72 [0.86, 3.44]	NA	NA	—	1	1.02 [0.40, 2.59]	NA	NA	—	1	1.01 [0.58, 1.78]	NA	NA	—
Neurotoxicity *(Pediatric)*	2	0.91 [0.36, 2.33]	0.89	0%	—	2	0.59 [0.24, 1.47]	0.5	0%	—	1	0.81 [0.36, 1.80]	NA	NA	—
GI toxicity *(Adult)*	1	0.75 [0.36, 1.57]	NA	NA	—	—	—	—	—	—	—	—	—	—	—
Overall toxicity *(Adult)*	1	0.86 [0.32, 2.31]	NA	NA	—	1	1.63 [0.43, 6.19]	NA	NA	—	1	1.02 [0.47, 2.24]	NA	NA	—
Therapeutic interference *(Pediatric)*	1	0.80 [0.29, 2.24]	NA	NA	—	1	0.60 [0.21, 1.77]	NA	NA	—	1	0.83 [0.39, 1.76]	NA	NA	—
EFS	2	Esmaili M A-2020: HR>1, P<0.05 Ma C X-2015: P=0.076													

Note: —: No available data

Abbreviation: a. NA: Not applicable; b. NS: No statistical difference

Table S6. Pooled results of the association between *FPGS (rs10106)* and HDMTX related outcomes

FPGS A1994G (rs10106)	MM/Mm vs mm （dominant model)					MM vs Mm/mm (recessive model)					M vs m (allelic model)				
	Studies	OR or HR (95%CI)	*Phet*	I^2	P-sub	Studies	OR or HR (95%CI)	*Phet*	I^2	P-sub	Studies	OR or HR (95%CI)	*Phet*	I^2	P-sub
Hepatotoxicity *(Adult)*	—	—	—	—	—	1	0.60 [0.27, 1.32]	NA [a]	NA	—	—	—	—	—	—

Note: —: No available data

Abbreviation: a. NA: Not applicable

Table S7. Meta-analysis results of the association between *FPGS (rs1544105)* and HDMTX related outcomes

FPGS G2752A (rs1544105)	MM/Mm vs mm （dominant model)					MM vs Mm/mm (recessive model)					M vs m (allelic model)				
	Studies	OR or HR (95%CI)	*Phet*	I^2	P-sub	Studies	OR or HR (95%CI)	*Phet*	I^2	P-sub	Studies	OR or HR (95%CI)	*Phet*	I^2	P-sub
2y-OS *(Adult)*	—	—	—	—	—	1	0.45 [0.24, 0.84]	NA [a]	NA	—	—	—	—	—	—

Note: —: No available data

Abbreviation: a. NA: Not applicable

Table S8. Pooled results of the association between *MTHFR (rs1801133)* and HDMTX related outcomes

MTHFR C677T (rs1801133)	MM/Mm vs mm （dominant model）					MM vs Mm/mm (recessive model)					M vs m (allelic model)				
	Studies	OR or HR (95%CI)	*Phet*	I^2	P-sub	Studies	OR or HR (95%CI)	*Phet*	I^2	P-sub	Studies	OR or HR (95%CI)	*Phet*	I^2	P-sub
Hepatotoxicity	**9**	**1.52 [1.03, 2.23]**	**0.01**	**60%**	**—**	**8**	**1.68 [1.10, 2.55]**	**0.08**	**45%**	**—**	**7**	**1.41 [1.01, 1.97]**	**0.69**	**0%**	**—**
Pediatric	6	1.19 [0.75, 1.89]	0.05	55%	NS [a]	5	0.80 [0.41, 1.54]	0.34	12%	**0.004***	4	1.02 [0.61, 1.71]	0.88	0%	NS
Adult	**3**	**2.62 [1.31, 5.25]**	**0.07**	**63%**		**3**	**2.76 [1.60, 4.74]**	**0.99**	**0%**		**3**	**1.79 [1.15, 2.77]**	**0.74**	**0%**	
G3-4 Hepatotoxicity	**3**	**0.16 [0.06, 0.41]**	**0.004**	**82%**	**—**	3	1.29 [0.57, 2.91]	0.08	45%	—	2	0.89 [0.47, 1.67]	0.83	0%	—
Pediatric	2	0.74 [0.15, 3.60]	0.42	0%	**0.001***	2	2.65 [0.56, 11.94]	0.59	0%	NS	1	0.74 [0.12, 4.47]	NA	NA	NS
Adult	**1**	**0.03 [0.01, 0.10]**	**NA [b]**	**NA**		1	0.96 [0.35, 2.63]	NA	NA		1	0.91 [0.46, 1.79]	NA	NA	
Renal toxicity	4	1.84 [0.92, 3.69]	0.07	58%	—	**4**	**3.54 [1.81, 6.90]**	**0.1**	**53%**	**—**	**4**	**1.89 [1.18, 3.02]**	**0.38**	**3%**	**—**
Pediatric	1	0.63 [0.24, 1.69]	NA	NA	**0.01***	1	2.36 [0.23, 24.26]	NA	NA	NS	1	1.67 [0.83, 3.36]	NA	NA	NS
Adult	**3**	**4.99 [1.48, 16.84]**	**0.96**	**0%**		**3**	**3.65 [1.81, 7.36]**	**0.04**	**68%**		**3**	**2.09 [1.11, 3.94]**	**0.24**	**31%**	
G3-4 Renal toxicity *(Pediatric)*	1	2.33 [0.22, 24.92]	NA	NA	—	—	—	—	—	—	—	—	—	—	—
Mucositis	**9**	**2.11 [1.31, 3.41]**	**<0.00001**	**79%**	**—**	9	1.52 [0.95, 2.41]	0.07	46%	—	**8**	**1.91 [1.28, 2.85]**	**<0.00001**	**82%**	**—**
Pediatric	**5**	**4.12 [2.28, 7.43]**	**0.00**	**78%**	**0.0002***	**5**	**2.46 [1.29, 4.71]**	**0.61**	**0%**	NS	**4**	**4.60 [2.58, 8.20]**	**0.0009**	**82%**	**0.0001***
Adult	3	0.49 [0.21, 1.14]	0.18	42%		3	0.79 [0.38, 1.66]	0.03	71%		3	0.75 [0.41, 1.37]	0.14	49%	
Mixed	1	7.86 [0.34, 180.33]	NA	NA		1	1.67 [0.37, 7.61]	NA	NA		1	1.60 [0.40, 6.36]	NA	NA	
G3-4 Mucositis *(Pediatric)*	3	1.23 [0.53, 2.85]	0.29	19%	—	1	0.52 [0.02, 11.03]	NA	NA	—	1	0.35 [0.04, 3.37]	NA	NA	—
GI toxicity	3	0.82 [0.36, 1.87]	0.57	0%	—	4	1.37 [0.59, 3.16]	0.38	2%	—	3	0.89 [0.39, 2.06]	0.67	0%	—
Pediatric	2	0.63 [0.25, 1.61]	0.99	0%	NS	2	0.91 [0.28, 2.92]	0.78	0%	NS	2	0.60 [0.18, 2.02]	0.96	0%	NS
Adult	1	2.20 [0.36, 1.87]	NA	NA		2	2.11 [0.64, 7.03]	0.16	50%		1	1.28 [0.40, 4.04]	NA	NA	
G3-4 GI toxicity	1	2.51 [0.10, 66.20]	NA	NA	—	1	1.73 [0.06, 48.17]	NA	NA	—	1	1.56 [0.09, 26.47]	NA	NA	—

MTHFR C677T *(rs1801133)*	MM/Mm vs mm （dominant model)					MM vs Mm/mm (recessive model)					M vs m (allelic model)				
	Studies	**OR or HR (95%CI)**	***Phet***	**I^2**	**P-sub**	**Studies**	**OR or HR (95%CI)**	***Phet***	**I^2**	**P-sub**	**Studies**	**OR or HR (95%CI)**	***Phet***	**I^2**	**P-sub**
(Pediatric)															
Dermal toxicity *(Pediatric)*	1	7.82 [0.45, 135.41]	NA	NA	—	1	1.93 [0.58, 6.43]	NA	NA	—	1	1.64 [0.62, 4.32]	NA	NA	—
Neurotoxicity *(Pediatric)*	2	0.81 [0.38, 1.73]	0.14	55%	—	3	0.94 [0.29, 3.10]	0.25	29%	—	1	0.40 [0.15, 1.06]	NA	NA	—
G3-4 Neurotoxicity *(Pediatric)*	2	0.90 [0.24, 3.40]	0.53	0%	—	1	5.38 [0.30, 95.06]	NA	NA	—	1	0.90 [0.21, 3.77]	NA	NA	—
Overall toxicity *(Adult)*	1	0.57 [0.21, 1.58]	NA	NA	—	1	0.85 [0.16, 4.55]	NA	NA	—	1	0.82 [0.38, 1.76]	NA	NA	—
Therapeutic interference *(Pediatric)*	**1**	**3.40 [1.31, 8.87]**	**NA**	**NA**	**—**	1	3.79 [0.92, 15.68]	NA	NA	—	1	2.24 [0.97, 5.16]	NA	NA	—
Relapse/Death *(Pediatric)*	2	0.81 [0.44, 1.52]	0.77	0%	—	**1**	**3.08 [1.38, 6.88]**	**NA**	**NA**	**—**	1	1.17 [0.65, 2.10]	NA	NA	—
5y-EFS *(Pediatric)*	2	0.83 [0.56, 1.25]	0.56	0%	—	1	0.63 [0.29, 1.35]	NA	NA	—	2	Yazicioglu B-2017: CT vs. TT vs. CC: P=0.72 Seidemann-2006: CC vs. CT vs. TT: p>0.05			
RFS *(Pediatric)*	1	0.70 [0.32, 1.54]	NA	NA	—	—	—	—	—	—	—	—	—	—	—
OS *(Pediatric)*	1	1.10 [0.47, 2.57]	NA	NA	—	—	—	—	—	—	—	—	—	—	—

Note: —: No available data

Abbreviation: a. NS: No statistical difference; b. NA: Not applicable

Table S9. Pooled results of the association between *MTHFR (rs1801131)* and HDMTX related outcomes

MTHFR A1298C (rs1801131)	MM/Mm vs mm （dominant model)					MM vs Mm/mm (recessive model)					M vs m (allelic model)				
	Studies	OR or HR (95%CI)	*Phet*	I^2	P-sub	Studies	OR or HR (95%CI)	*Phet*	I^2	P-sub	Studies	OR or HR (95%CI)	*Phet*	I^2	P-sub
Hepatotoxicity	8	0.70 [0.47, 1.03]	0.29	18%	—	7	1.12 [0.61, 2.05]	0.78	0%	—	5	0.88 [0.56, 1.37]	0.94	0%	—
Pediatric	6	0.59 [0.37, 0.92]	0.73	0%	NS [a]	5	1.29 [0.59, 2.82]	0.57	0%	NS	3	0.75 [0.41, 1.39]	0.92	0%	NS
Adult	2	1.20 [0.54, 2.66]	0.6	0%		2	0.92 [0.36, 2.35]	0.98	0%		2	1.05 [0.55, 1.99]	0.82	0%	
G3-4 Hepatotoxicity *(Pediatric)*	2	2.89 [0.44, 19.03]	0.86	0%	—	1	2.17 [0.18, 26.29]	NA [b]	NA	—	1	1.63 [0.36, 7.43]	NA	NA	—
Renal toxicity	3	0.61 [0.32, 1.13]	0.76	0%	—	2	0.79 [0.14, 4.40]	0.07	70%	—	**2**	**0.41 [0.18, 0.97]**	**0.59**	**0%**	**—**
Pediatric	2	0.53 [0.26, 1.11]	0.75	0%	NS	1	5.94 [0.54, 66.05]	NA	NA	NS	1	0.48 [0.18, 1.28]	NA	NA	NS
Adult	1	0.87 [0.25, 3.04]	NA	NA		1	0.23 [0.01, 4.43]	NA	NA		1	0.28 [0.06, 1.45]	NA	NA	
G3-4 Renal toxicity *(Pediatric)*	1	0.68 [0.09, 5.49]	NA	NA	—	—	—	—	—	—	—	—	—	—	—
Mucositis	7	0.77 [0.51, 1.17]	0.76	0%	—	6	0.70 [0.33, 1.47]	0.3	18%		4	0.73 [0.46, 1.15]	0.98	0%	—
Pediatric	6	0.85 [0.54, 1.35]	0.79	0%	NS	5	0.46 [0.17, 1.28]	0.32	16%	NS	3	0.69 [0.39, 1.21]	0.95	0%	NS
Adult	1	0.48 [0.18, 1.33]	NA	NA		1	1.13 [0.38, 3.35]	NA	NA		1	0.81 [0.37, 1.77]	NA	NA	
G3-4 Mucositis *(Pediatric)*	2	0.99 [0.18, 5.45]	0.54	0%	—	1	14.00 [1.23, 158.84]	NA	NA	—	1	1.56 [0.20, 12.05]	NA	NA	—
GI toxicity *(Pediatric)*	1	0.45 [0.02, 11.82]	NA	NA	—	3	0.54 [0.18, 1.64]	0.7	0%	—	1	0.66 [0.04, 11.12]	NA	NA	—
G3-4 GI toxicity *(Pediatric)*	1	2.23 [0.08, 58.81]	NA	NA	—	1	2.19 [0.08, 62.33]	NA	NA	—	1	1.53 [0.09, 25.90]	NA	NA	—
Dermal toxicity *(Pediatric)*	1	0.09 [0.01, 1.53]	NA	NA	—	1	1.23 [0.06, 24.20]	NA	NA	—	1	0.12 [0.01, 2.15]	NA	NA	—
Neurotoxicity *(Pediatric)*	1	1.42 [0.48, 4.22]	NA	NA	—	2	1.36 [0.34, 5.40]	0.43	0%	—	1	1.15 [0.51, 2.60]	NA	NA	—

MTHFR A1298C *(rs1801131)*	MM/Mm vs mm （dominant model)					MM vs Mm/mm (recessive model)					M vs m (allelic model)				
	Studies	**OR or HR (95%CI)**	***Phet***	**I^2**	**P-sub**	**Studies**	**OR or HR (95%CI)**	***Phet***	**I^2**	**P-sub**	**Studies**	**OR or HR (95%CI)**	***Phet***	**I^2**	**P-sub**
G3-4 Neurotoxicity *(Pediatric)*	1	Both are zero-events	NA	NA	—	—	—	—	—	—	—	—	—	—	—
Overall toxicity *(Adult)*	1	0.93 [0.36, 2.40]	NA	NA	—	1	2.74 [0.27, 27.69]	NA	NA	—	1	1.02 [0.45, 2.30]	NA	NA	—
Therapeutic interference *(Pediatric)*	1	2.03 [0.79, 5.18]	NA	NA	—	1	1.21 [0.35, 4.18]	NA	NA	—	1	1.41 [0.64, 3.10]	NA	NA	—
Relapse/Death *(Pediatric)*	2	0.69 [0.36, 1.32]	0.03	79%	—	1	0.78 [0.30, 2.01]	NA	NA	—	1	1.00 [0.58, 1.72]	NA	NA	—
EFS *(Pediatric)*	2	Erculj N-2012: 0.66 [0.33, 1.30] Fukushima H-2013: HR<1, P<0.05				—	—	—	—	—	2	Yazicioglu B-2017: AC vs. CC vs. AA *P=0.59* Radtke S-2013: AC vs AA *HR=1.2 [0.5-2.7] P=0.676* Radtke S-2013: CC vs AA *HR=3.1 [1.2-7.7] P=0.015*			
RFS *(Pediatric)*	1	0.50 [0.22, 1.14]	NA	NA	—	—	—	—	—	—	—	—			
OS *(Pediatric)*	1	0.63 [0.27, 1.45]	NA	NA	—	—	—	—	—	—	—	—	—	—	—

Note: —: No available data

Abbreviation: a. NS: No statistical difference; b. NA: Not applicable

Table S10. Pooled results of the association between *TYMS (rs34743033)* and HDMTX related outcomes

TYMS (rs34743033)	MM/Mm vs mm （dominant model)					MM vs Mm/mm (recessive model)					M vs m (allelic model)				
	Studies	OR or HR (95%CI)	*Phet*	I^2	P-sub	Studies	OR or HR (95%CI)	*Phet*	I^2	P-sub	Studies	OR or HR (95%CI)	*Phet*	I^2	P-sub
Hepatotoxicity *(Pediatric)*	3	0.60 [0.23, 1.59]	0.75	0%	—	—	—	—	—	—	—	—	—	—	—
Renal toxicity *(Pediatric)*	1	0.87 [0.22, 3.42]	NA [a]	NA	—	—	—	—	—	—	—	—	—	—	—
Mucositis *(Pediatric)*	6	0.66 [0.47, 0.94]	0.57	0%	—	1	0.90 [0.42, 1.90]	NA	NA	—	1	0.81 [0.40, 1.62]	NA	NA	—
G3-4 Mucositis *(Pediatric)*	2	0.86 [0.46, 1.60]	0.04	76%	—	1	0.95 [0.33, 2.72]	NA	NA	—	1	1.19 [0.56, 2.52]	NA	NA	—
Neurotoxicity *(Pediatric)*	2	2.94 [0.55, 15.83]	0.32	0%	—	1	2.16 [0.73, 6.42]	NA	NA	—	1	1.91 [0.67, 5.39]	NA	NA	—
Overall toxicity *(Pediatric)*	1	0.93 [0.32, 2.74]	NA	NA	—	—	—	—	—	—	—	—	—	—	—
RFS *(Pediatric)*	1	0.56 [0.24, 1.32]	NA	NA	—	—	—	—	—	—	—	—	—	—	—
OS *(Pediatric)*	1	0.59 [0.24, 1.45]	NA	NA	—	—	—	—	—	—	—	—	—	—	—
EFS *(Pediatric)*	1	0.54 [0.26, 1.13]	NA	NA	—	—	—	—	—	—	1	Yazicioglu B-2017: 2R3R vs. 3R3R vs. 2R2R: P=0.15			

Note: —: No available data

Abbreviation: a. NA: Not applicable

Table S11. Pooled results of the association between *ATIC (rs2372536)* and HDMTX related outcomes

ATIC C347G (rs2372536)	MM/Mm vs mm （dominant model)					MM vs Mm/mm (recessive model)					M vs m (allelic model)				
	Studies	OR or HR (95%CI)	*Phet*	I^2	P-sub	Studies	OR or HR (95%CI)	*Phet*	I^2	P-sub	Studies	OR or HR (95%CI)	*Phet*	I^2	P-sub
Neurotoxicity	1	1.36 [0.46, 4.06]	NA [a]	NA	—	1	1.74 [0.23, 13.40]	NA [a]	NA	—	—	—	—	—	—

Note: —: No available data

Abbreviation: a. NA: Not applicable

Supplemental Material V

Figure S1. Forest plot for meta-analysis of the association between *RFC1 (rs1051266)* and hepatotoxicity under recessive model

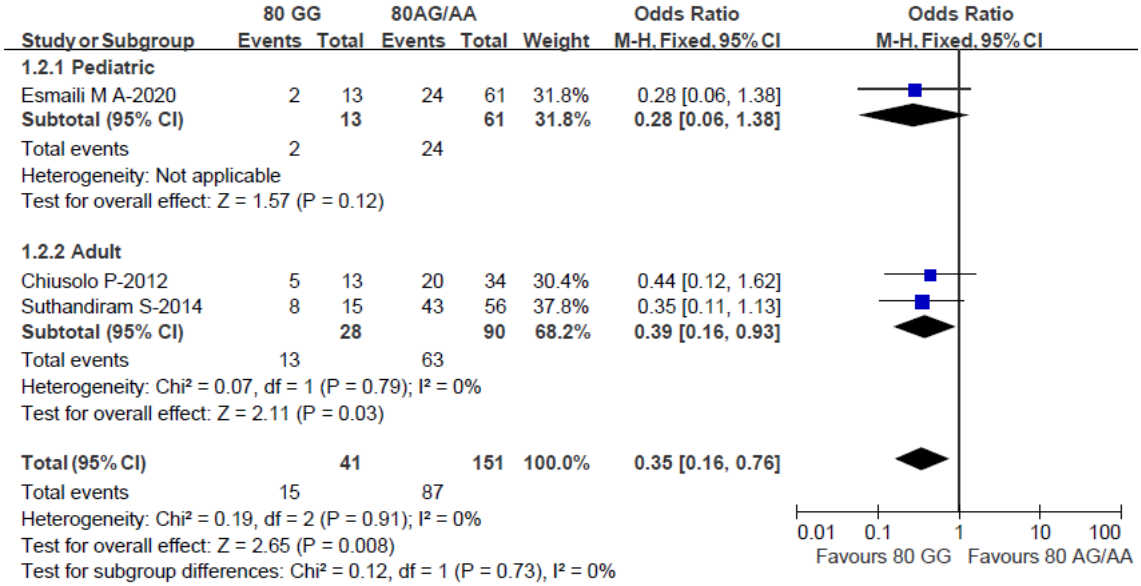


Figure S2. Forest plot for meta-analysis of the association between *ABCB1* (rs1045642) and hepatotoxicity under dominant model

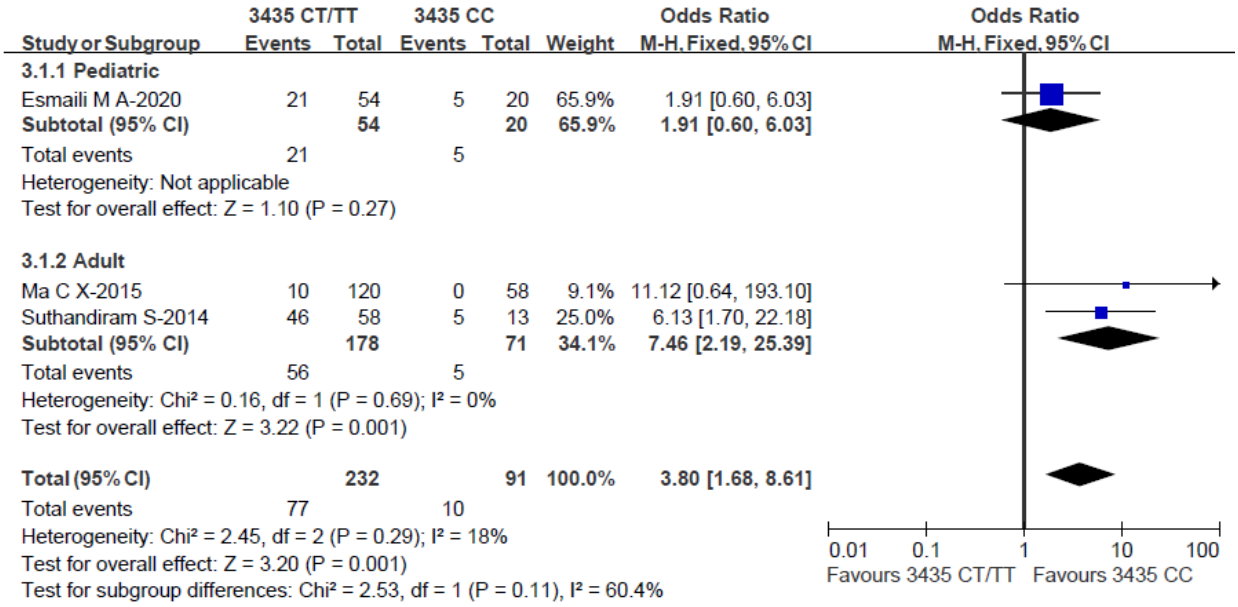


Figure S3. Forest plot for meta-analysis of the association between *MTHFR (rs1801133)* and hepatotoxicity under dominant model

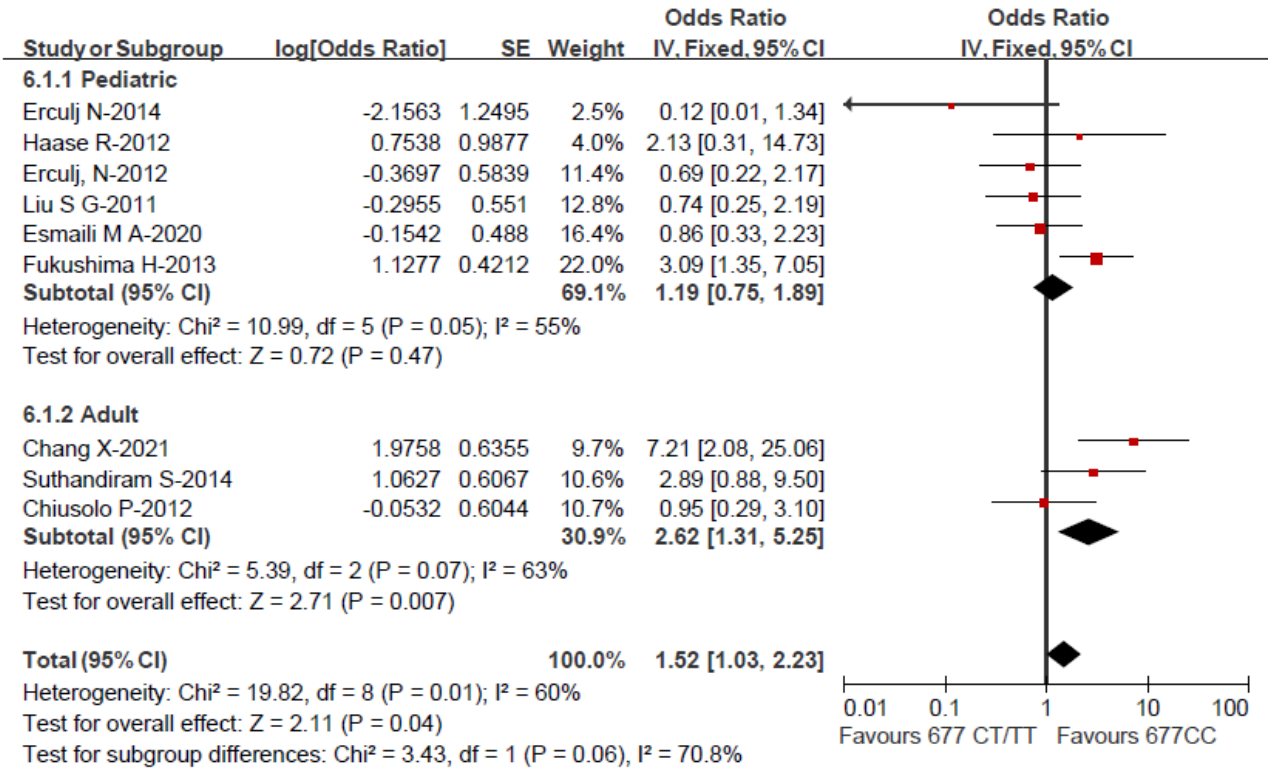
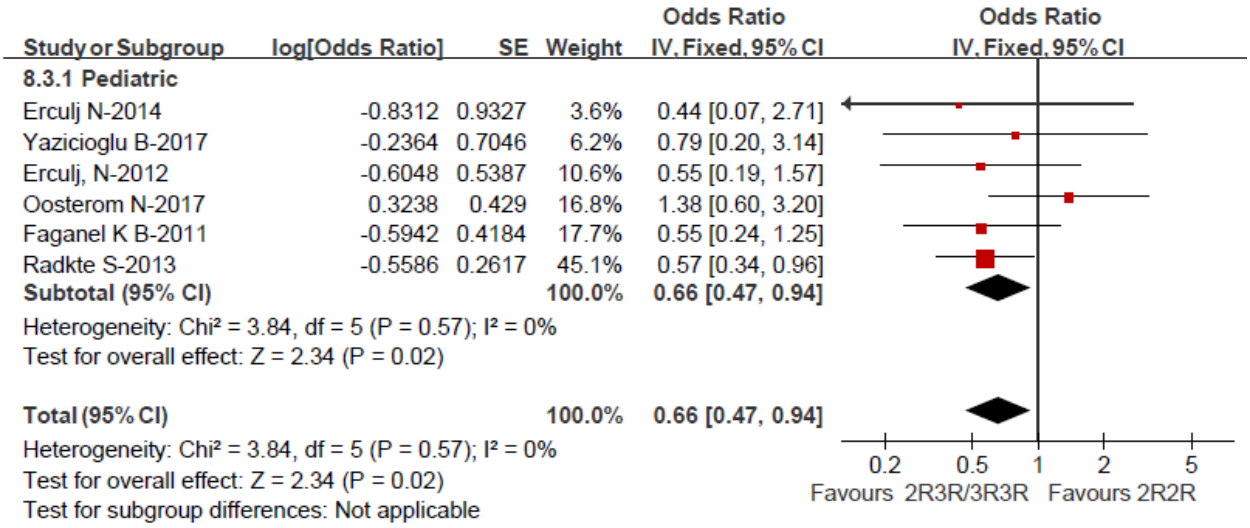


Figure S4. Forest plot for meta-analysis of the association between *TYMS (rs34743033)* and mucositis under dominant model



Supplemental Material VI

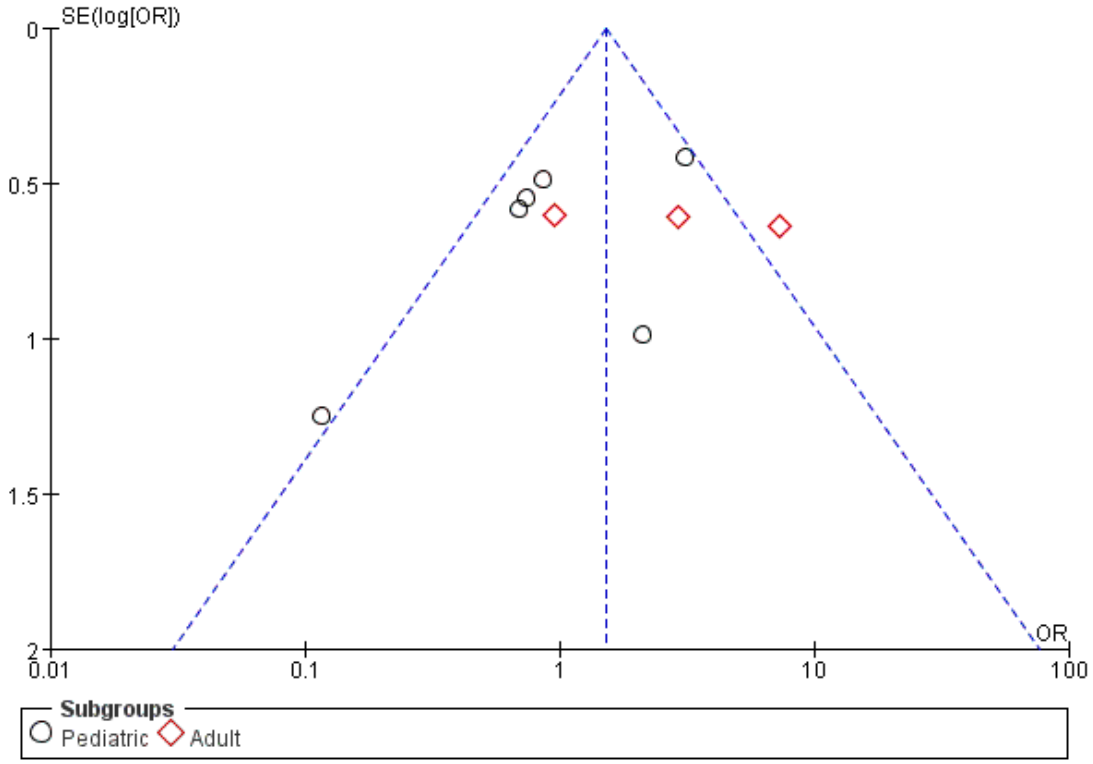



Figure S5. Funnel plot for publication bias test of the association between *MTHFR (rs1801133)* and hepatotoxicity under dominant model